

Day 5: Network Testing Practical

GitHub-Ready Assignment Report

Objective

To test network connectivity using ping, tracert and arp -a.

Tools Used: Windows laptop, Wi-Fi network, smartphone, Command Prompt.

Task 1: Test Connectivity Using Ping

The laptop and phone were connected to the same Wi-Fi. The phone IP was checked and ping was used from the laptop.

```
ping 192.168.1.50
```

Result: Reply received, showing that both devices could communicate.

Screenshot: Add ping output here.

Task 2: Run Traceroute / Tracert

Windows uses tracert to show the path to a website.

```
tracert www.youtube.com
```

Result: Sample test showed about 12 hops.

Screenshot: Add tracert output here.

Task 3: Test After Disconnecting

The phone was disconnected from Wi-Fi and ping was run again.

```
ping 192.168.1.50
```

Result: Request timed out because the phone was offline.

Task 4: Check Devices Using ARP

The arp -a command was used to view IP and MAC addresses known by the laptop.

```
arp -a
```

IP Address	MAC Address	Device	Status
192.168.1.1	AA-BB-CC-11-22-33	Wi-Fi Router	Active
192.168.1.20	AA-BB-CC-44-55-66	My Laptop	Active
192.168.1.50	AA-BB-CC-77-88-99	Friend's Phone	Active

Note: Replace sample IP/MAC addresses with your actual network values.

Screenshot: Add arp -a output here.

Task 5: Network Testing Report

I connected my laptop and phone to the same Wi-Fi and tested the connection using ping. The laptop received replies from the phone. I then used tracert to check the path to YouTube and observed about 12 hops. After disconnecting the phone, ping showed a timeout. Finally, arp -a was used to view connected device IP and MAC addresses.

Conclusion

The practical helped me understand basic network connectivity, traceroute and device identification.

Add actual screenshots before uploading the PDF to GitHub.